

Mustafa Talha Avcu

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WORK EXPERIENCE

- Amazon** Seattle, WA & Toronto, ON
 - Software Development Engineer II* *August 2022 - June 2026*
 - Implemented a guardrail layer within a new LLM orchestration service connecting generative AI models with Amazon's customer service chatbot. Validated customer inputs and model outputs to ensure safe and appropriate interactions, contributing to the launch of Amazon's first LLM-powered customer support experience.
 - Designed and built reusable ML infrastructure using AWS CodePipeline, S3, and SageMaker to automate model training, testing, and deployment through a CI/CD pipeline. Deployed a model using this infrastructure to infer customer issues and route them to relevant support workflows, reducing average chatbot interaction time by 9% (440s to 400s) with no negative impact on other metrics.
 - Owned and implemented the integration of Amazon Demand-Side Platform (ADSP) into the Retailer Ad Service's attribution service, extending existing Sponsored Products attribution to also support display and video advertising. Built unified data pipelines and attribution logic across both advertising programs.
 - Led scaling efforts for the Retailer Ad Service catalog and ingestion services to support Macy's onboarding, handling catalogs with 2M+ offers uploaded as frequently as every two hours. Improved ingestion scalability and reliability, resolving service bottlenecks and enabling stable onboarding of the platform's largest retailer to date.
- Agency for Science, Technology and Research (A*STAR)** Singapore
 - Data Science Intern* *Jun 2018 - Sept 2018*
 - Developed **SeizNet**, a CNN-based seizure detection model, achieving 95.8% sensitivity with 0.17 false alarms/hour on EEG data from 29 patients; demonstrated that a 2-channel configuration achieved 93.3% sensitivity, outperforming a traditional SVM baseline using the full set of EEG channels.

TECHNICAL SKILLS

Programming Languages: Java, Python, Scala, JavaScript, TypeScript, SQL, C++

Machine Learning and Data: Spark, Large Language Models (LLMs), Keras, TensorFlow, Scikit-learn

AWS Technologies: AWS CDK, CloudFormation, CodePipeline, S3, Lambda, ECS, SQS/SNS, CloudWatch, Clean Rooms, SageMaker, Bedrock

Others: Linux, Docker

EDUCATION

- University of Southern California** Los Angeles, CA
 - Master of Science in Computer Science, CGPA: 3.96/4.00* *May 2022*
 - Selected Courses: Machine Learning, Analysis of Algorithms, Database Systems, Applied Machine Learning for Games, Estimation Theory, Random Processes, Applied Natural Language Processing
- Bilkent University** Ankara, Turkey
 - Bachelor of Science in Electrical and Electronics Engineering, CGPA: 3.93/4.00* *Jun 2019*

HONORS & AWARDS

- Two-time Amazon org-wide hackathon winner among 20+ competing teams**
- Annenberg Fellowship with admission to University of Southern California (USC) PhD program, 2019**
- IEEEExtreme Programming Competition**
 - IEEEExtreme 10.0, 2016:** Ranked 2nd in Turkey, 20th in IEEE Region 8, and 65th overall
 - IEEEExtreme 9.0, 2015:** Ranked 3rd in Turkey, 30th in IEEE Region 8, and 82nd overall
- Nationwide University Entrance Exam (LYS):** Ranked in top 0.01% among 2 million students in Turkey, 2014

PUBLICATIONS

M. T. Avcu, Z. Zhang, and DWS. Chan "Seizure Detection Using Least EEG Channels by Deep Convolutional Neural Networks" IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), 2019